

Improving Visual Sensitivity of LLMs on Multimodal Machine Translation with Metric-based Loss Weighting

Paweł Mąka¹, Piotr Andruszkiewicz^{2,3}, Yusuf Can Semerci¹, Jan Scholtes¹,
Gerasimos Spanakis¹

¹Department of Advanced Computing Sciences, Maastricht University

²Warsaw University of Technology

³IDEAS Research Institute

{pawel.maka, y.semerci, j.scholtes, jerry.spanakis}@maastrichtuniversity.nl
piotr.andruszkiewicz@pw.edu.pl

Abstract

Multimodal Machine Translation aims to incorporate additional signal from non-textual modalities to improve translations by resolving ambiguities. While models, through multimodal fusion, are able to accept images related to the source text, they can ignore this information. Therefore, increasing their visual sensitivity remains an active research area. In this work, we introduce a training method, Metric-based Loss Weighting, that improves visual grounding of translations by increasing the loss function for tokens that benefit from the accompanying image. We identify these tokens using the Point-wise Cross-mutual Information (PCXMI) metric, which compares the model’s output probabilities with and without visual context. We introduce a Congruency-based PCXMI metric and experimentally show that both metrics working in combination yield the best results. We evaluate our method by fine-tuning three pretrained Multimodal Large Language Models on the task of Image-guided Machine Translation for three language directions. Metric-based Loss Weighting outperforms other tested methods on the CoMMuTE contrastive dataset, improving accuracy by up to more than 7 percentage points compared to standard fine-tuning, while maintaining strong general translation performance.

1 Introduction

Sentence-level Machine Translation (MT) systems achieved high performance (Vaswani et al., 2017; Hassan et al., 2018; Costa-jussà et al., 2022; Tiedemann et al., 2024), but they are inherently limited by the lack of context. In Multimodal Machine Translation (MMT), the additional signal from other modalities can improve translation in cases where text alone is ambiguous. This signal can be audio, video, or image (Sulubacak et al., 2020; Garg et al., 2022; Feng et al., 2025). In this

study, we concentrate our efforts on Image-based MT.

While the images are available to the MMT systems, the reliance on the other modality is not guaranteed. Indeed, Elliott (2018) did not find the expected performance drop of the tested systems after randomly replacing images. Wu et al. (2021) attributed some gains of the MMT systems to regularization. Therefore, visual sensitivity of the MMT models has seen increased attention from researchers in recent years (Sato et al., 2023; Zhu et al., 2023; Cheng et al., 2024; Bowen et al., 2024; Futral et al., 2025).

As traditional translation metrics such as BLEU (Papineni et al., 2002) are not sensitive enough to judge disambiguations, we turn towards CoMMuTE (Futral et al., 2023). It is a contrastive dataset where the tested models are presented with multiple possible translations of the ambiguous source sentence and tasked with selecting the correct one. The selection is based on the probability that models give to each option, and the final metric is the accuracy of assigning the highest probability to the correct one.

The field of MMT went from training specialized architectures from random initialization (Yin et al., 2020; Liu et al., 2021) to using pre-trained models as building blocks and fine-tuning the resulting model with multimodal parallel data (Gupta et al., 2023; Yu et al., 2025). Recently, more focus has been given to Multimodal Large Language Models (MLLMs) (Alayrac et al., 2022; Liu et al., 2023; Laurençon et al., 2023; Bai et al., 2025) due to their generalizability and strong performance in different tasks (Zhu et al., 2024). In this study, we concentrate on methods that could be used to enhance fine-tuning of MLLMs on the task of MMT to increase the visual grounding of the translation.

We argue that the insufficient number of exam-

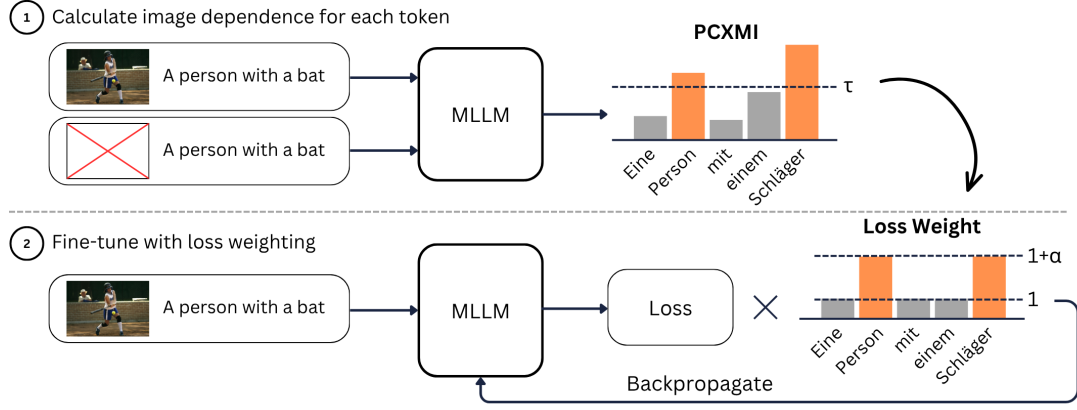


Figure 1: The illustration of Metric-based Loss Weighting: (1) Calculate the image dependence metric for each token (PCXMI), based on the model’s output probabilities with and without an image (or with an incongruent image); (2) Fine-tune the model with increased loss (by α) for the identified tokens (with PCXMI larger than τ).

ples in the widely used MMT training datasets that require images for correct translation is causing the training signal to be sparse and, therefore, not enough for the models to learn to rely on context (Futeral et al., 2023). To mitigate this issue, we propose a training strategy where we identify the tokens in the target sequence that benefit from context and, subsequently, increase the loss for those tokens during training. To identify the image-dependent tokens, we use the Point-wise Cross-Mutual Information (PCXMI) metric (Fernandes et al., 2023), which compares the output probabilities of the model in two situations: with and without the context (see Figure 1). We propose a modification to this metric by using an incongruent image as context instead of the lack of it. We combine both metrics to achieve the best results. We test our method against several techniques adapted to fine-tuning of MLLMs using three models and three language directions. Our Metric-based Loss Weighting method outperforms the baselines in terms of CoMMuTE accuracy on most of the tested scenarios. It improves the accuracy by up to more than 7 percentage points compared to the standard fine-tuning.

Our contributions are as follows:

1. We introduce a training strategy for fine-tuning MLLMs that improves their visual sensitivity on the task of MMT based on identifying and increasing loss function for image-dependent tokens.
2. We adapt the Point-wise Cross-Mutual Information (PCXMI) metric (Fernandes et al., 2023) to MMT, introduce Congruency-based PCXMI metric, and experimentally show that both metrics working in combination yield best results.

3. We adapt and evaluate several existing methods for fine-tuning MLLMs on the task of MMT, establishing the baseline for future research.

4. We show that the pre-trained models exhibit high visual sensitivity which can be further increased with specialized fine-tuning techniques.

5. We empirically confirm the discrepancy between traditional translation metrics such as BLEU (Papineni et al., 2002) and the image reliance of the MLLMs.

2 Related Work

Researchers have been interested in incorporating the information coming from other modalities to enhance text-to-text translation (Sulubacak et al., 2020; Garg et al., 2022; Feng et al., 2025). While this includes audio and video (Gu et al., 2021; Li et al., 2023), large focus has been concentrated on single images related to the source text (Elliott et al., 2017; Barrault et al., 2018; Cheng et al., 2024; Futeral et al., 2025; Yu et al., 2025). The methods proposed specialized architectures concentrating on feature extraction (Yao and Wan, 2020; Fang and Feng, 2022), cross-modal fusion and alignment (Zhao et al., 2022; Guo et al., 2022), and cross-modal attention (Nishihara et al., 2020; Gupta et al., 2023).

Other notable directions include: employing auxiliary tasks, such as Visually Conditioned Masked Language Modeling (V MLM) (Futeral et al., 2023, 2025), image captioning (Gupta et al., 2023), or visual question answering (Zhang et al., 2025; Gao et al., 2025), masking the source sentence to increase visual grounding (Sato et al., 2023; Cheng

et al., 2024; Bowen et al., 2024), and data augmentation by image generation (Long et al., 2021; Li et al., 2022c; Chen et al., 2025), using text-to-image models (Goodfellow et al., 2014; Rombach et al., 2022), or backtranslation (Li et al., 2021).

The MMT models are often trained on the widely used Multi30k dataset and evaluated on the accompanying testsets (Elliott et al., 2016, 2017; Barrault et al., 2018). Although the systems show increased BLEU scores (Papineni et al., 2002) compared to the text-only baselines, the benefits have been attributed to regularization (Wu et al., 2021). For example, Elliott (2018), after replacing the relevant images with random ones, did not see the expected drop in performance. This could be caused by the insufficient number of examples in the Multi30k training dataset that require context in the form of the accompanying image for correct translation (Futeral et al., 2023).

Earlier works trained models from random initialization (Yin et al., 2020; Liu et al., 2021), but increasingly often researchers utilize pretrained models (Gupta et al., 2023; Futeral et al., 2025; Yu et al., 2025). This transition is motivated by the strong generalization capabilities of the pretrained models, acquired through training on large amounts of data. The popular models include CLIP (Radford et al., 2021) and BLIP (Li et al., 2022b) for image feature extraction, and Large Language Models, such as mBART (Liu et al., 2020) or Qwen (Yang et al., 2025), for text understanding and generation.

The emergence of strong general-purpose Multimodal Large Language Models (MLLMs), capable of incorporating visual information without architectural modifications, presents a potential shift in the approach to Multimodal Machine Translation (Alayrac et al., 2022; Awadalla et al., 2023; Bai et al., 2025; Chen et al., 2024). While Xiong and Zhao (2025) and Yu et al. (2025) evaluated their methods against MLLMs, and Zhu et al. (2024) performed a large scale evaluation of publicly-available MLLMs, we concentrate on instruction-tuning (Wei et al., 2022; Zhang et al., 2026; Hu et al., 2022) and evaluating the open-weights models (Laurençon et al., 2023; Bai et al., 2025).

Our method is inspired by Metric-based Fine-tuning and Loss Weighting, the two separate techniques introduced for encoder-decoder models by Mąka et al. (2025) for the task of Context-aware MT. Importantly, both techniques used only a textual modality, and the loss weighting was based on the external annotations. In this work, we combine

both techniques, and adapt it to the multimodal setting of Image-guided MT. Furthermore, we introduce a new Metric that uses incongruent images instead of not providing an image.

3 Method

Our proposed method consists of two steps. First, we identify which tokens in the target sentence require context. To do this, we measure how much the output probabilities of the pretrained model change when the context image is present or not (alternatively, when the image is not related to the text). Second, we use this information during training to improve the visual sensitivity of the model by increasing the loss corresponding to the context-dependent tokens.

Similar to Mąka et al. (2025), we employ the Point-wise Cross-Mutual Information (PCXMI) metric Fernandes et al. (2023) for the identification step. The original PCXMI metric compares the model’s output probabilities with and without previous sentences as context. We adapt it to the setting of Image-guided MT by treating the accompanying image as context. The equation below calculates the *PCXMI* for the j -th token in the target sequence:

$$PCXMI_j = \log \frac{q(y_j|y_{t<j}, x, C)}{q(y_j|y_{t<j}, x)}, \quad (1)$$

where $q(y_j|y_{t<j}, x, C)$ is the probability of the j -th token conditioned on the previous tokens $y_{t<j}$, source sentence x , and the image C , and $q(y_j|y_{t<j}, x)$ is the probability conditioned only on the previous tokens $y_{t<j}$ and the source sentence x . Crucially, the model has to be able to perform translation both with and without the image to use this metric.

Additionally, we propose a modification to the metric, where the comparison is between the probabilities with the correct and incorrect (incongruent) images. We refer to this metric as Congruency-based Point-wise Cross-Mutual Information (CPCMXI):

$$CPCMXI_j = \log \frac{q(y_j|y_{t<j}, x, C)}{q(y_j|y_{t<j}, x, \tilde{C})}, \quad (2)$$

where $q(y_j|y_{t<j}, x, \tilde{C})$ is the probability conditioned on the previous tokens $y_{t<j}$, source sentence x , and incongruent image $\tilde{C}$, which we randomly sample from the training dataset. The advantage

over the standard PCXMI is that it can be used with models that always require the image to be present.

In practice, both metrics rely on the outputs of the model in different situations and are, therefore, noisy. To mitigate this, we calculate both metrics for each example d_i in the training dataset $\mathcal{D}$, and take the maximum value for each token as the final image dependence metric:

$$a_{i,j} = \max(PCXMI_{i,j}, CPCXMI_{i,j}). \quad (3)$$

Taking the maximum increases the reliability of the metric in situations where one or the other metric fails to mark a token as context-dependent.

In the second step, we apply token-level weighting of the standard Negative Log Likelihood loss based on the calculated context dependence metric (Mała et al., 2025):

$$\mathcal{L} = -\frac{1}{|D|} \sum_{(d_i, a_i) \sim D} \frac{1}{|y_i|} \sum_{j=1}^{|y_i|} w(a_{i,j}) \log(\hat{y}_{i,j}), \quad (4)$$

where d_i is the i -th example sampled from the training dataset $\mathcal{D}$ with the calculated metric a_i for each token in the target sentence, $\hat{y}_{i,j}$ is the output probability of the model for the j -th token, and $w(a_{i,j})$ is the weight function based on the value of the metric $a_{i,j}$ for the j -th token. We use the threshold function to obtain the weighting values:

$$w(a_{i,j}) = \begin{cases} 1 + \alpha, & \text{if } a_{i,j} > \tau, \\ 1, & \text{otherwise,} \end{cases} \quad (5)$$

where the hyper-parameters α and τ control the loss increase and the threshold, respectively.

Intuitively, we penalize the model more for the mistakes in tokens that require context. It should be noted that our method relies on the identification of the image-dependent tokens, which is done by the model itself. This means that we bias the model towards the tokens for which it already exhibits strong visual sensitivity.

4 Experiments

We experimentally evaluated the effects of our method on fine-tuning MLLMs on the task of MMT. We are specifically interested in observing how it affects the visual sensitivity of the tested models.

We experimented with three popular open-weights instruction-tuned models: Qwen3-

VL 4B Instruct¹, Qwen3-VL 8B Instruct² (Bai et al., 2025; Yang et al., 2025), and IDEFICS 9B Instruct³ (Laurençon et al., 2023). We used LoRA (Hu et al., 2022) to fine-tune the models on the Multi30k dataset (Elliott et al., 2016). We experimented with three language directions: English-to-German, English-to-French, and English-to-Czech.

We compared our method with the following baselines:

- **Fine-tuning** - simply fine-tuning the model on the Image-guided MT training dataset.

- **Visually Conditioned Masked Language Modeling (VMLM)** - fine-tuning with the auxiliary task, where the model is presented with the source sentence containing masked tokens, and it has to predict those masked tokens (Futeral et al., 2023, 2025). We mask words with the probability p , which is a hyper-parameter, with a string MASK{n}, where n goes from 1 to the total number of masked words in the source text. Because VMLM does not require a parallel corpus, often an additional multimodal dataset in the source language is used. In our setting, we use only the Multi30k dataset, discarding the target-side sentences, to keep the methods using comparable amounts of data.

- **Metric-based Fine-tuning** - we adapt this method from the original text-only version (Mała et al., 2025). After the initial fine-tuning, we calculate the PCXMI and CPCXMI metrics and select top- k examples with the highest maximum values of the calculated metrics for further fine-tuning. We take the maximum of the two metrics, as this yielded better results in the preliminary experiments than either of the metrics on their own.

- **Word Dropout** - dropping random tokens from the source sentence to force the model to use the image to improve translation, adapted to Image-guided Machine Translation from CoWord Dropout (Fernandes et al., 2021). The probability of dropping out a token is controlled by the hyper-parameter p . In the preliminary experiments, we determined that dropout on the word-level was better than token-level. Therefore, we drop out words based on the Moses Tokenizer⁴ (Koehn et al., 2007).

¹<https://huggingface.co/Qwen/Qwen3-VL-4B-Instruct>

²<https://huggingface.co/Qwen/Qwen3-VL-8B-Instruct>

³<https://huggingface.co/HuggingFaceM4/idefics-9b-instruct>

⁴We used the Sacremoses implementation <https://>

Model	test 2016		test 2017		MSCOCO		Average		CoMMuTE Accuracy
	BLEU	COMET	BLEU	COMET	BLEU	COMET	BLEU	COMET	
Qwen3-VL 4B	38.51	85.56	32.95	84.90	29.94	81.24	33.80	83.90	74.00%
Fine-tuning	42.61 ± 0.12	86.66 ± 0.06	38.08 ± 0.38	85.98 ± 0.06	35.22 ± 0.45	82.70 ± 0.13	38.64 ± 0.18	85.11 ± 0.05	71.67 $\pm 0.19\%$
VMLM	42.52 ± 0.19	<u>86.79</u> ± 0.03	38.48 ± 0.43	86.34 ± 0.06	36.47 ± 0.09	<u>82.95</u> ± 0.17	39.16 ± 0.06	85.36 ± 0.05	71.67 $\pm 0.38\%$
Metric Fine-tuning	42.57 ± 0.22	86.47 ± 0.13	38.43 ± 0.10	85.99 ± 0.12	36.18 ± 0.16	82.86 ± 0.18	39.06 ± 0.15	85.10 ± 0.11	72.89 $\pm 0.44\%$
Word Dropout	42.20 ± 0.19	86.62 ± 0.06	37.17 ± 0.19	85.80 ± 0.01	35.00 ± 0.22	82.69 ± 0.16	38.13 ± 0.18	85.03 ± 0.04	76.89 $\pm 0.44\%$
Degradation	42.18 ± 0.16	86.68 ± 0.00	37.89 ± 0.03	85.88 ± 0.02	35.66 ± 0.26	82.85 ± 0.04	38.58 ± 0.14	85.14 ± 0.02	73.44 $\pm 0.59\%$
Metric Weighting	41.91 ± 0.33	86.80 ± 0.04	37.29 ± 0.14	<u>86.14</u> ± 0.04	35.71 ± 0.19	83.09 ± 0.13	38.30 ± 0.15	<u>85.34</u> ± 0.07	77.00 $\pm 0.33\%$
Qwen3-VL 8B	38.24	85.64	34.24	85.77	31.78	82.25	34.76	84.55	75.33%
Fine-tuning	42.87 ± 0.54	86.68 ± 0.21	38.15 ± 0.29	86.35 ± 0.10	35.93 ± 0.22	83.11 ± 0.32	38.98 ± 0.26	85.38 ± 0.19	69.33 $\pm 0.19\%$
VMLM	42.25 ± 0.92	86.84 ± 0.15	37.24 ± 0.62	86.09 ± 0.10	35.28 ± 1.35	83.33 ± 0.21	38.26 ± 0.96	85.42 ± 0.14	69.78 $\pm 0.97\%$
Metric Fine-tuning	43.15 ± 0.16	86.80 ± 0.10	38.19 ± 0.41	86.45 ± 0.07	36.19 ± 0.34	83.41 ± 0.12	39.18 ± 0.28	85.55 ± 0.08	73.56 $\pm 0.62\%$
Word Dropout	43.34 ± 0.25	87.17 ± 0.03	37.08 ± 0.18	86.26 ± 0.09	35.46 ± 0.74	83.55 ± 0.11	38.63 ± 0.18	85.66 ± 0.04	76.00 $\pm 0.33\%$
Degradation	43.47 ± 0.34	<u>87.01</u> ± 0.04	38.38 ± 0.14	86.55 ± 0.07	36.42 ± 0.19	83.34 ± 0.16	39.43 ± 0.10	<u>85.63</u> ± 0.08	72.22 $\pm 0.29\%$
Metric Weighting	42.82 ± 0.13	86.86 ± 0.11	37.38 ± 0.30	<u>86.47</u> ± 0.07	35.12 ± 0.31	83.31 ± 0.15	38.44 ± 0.14	85.55 ± 0.06	77.22 $\pm 0.22\%$
IDEFICS 9B	27.07	80.04	22.51	78.99	19.35	74.38	22.98	77.80	68.67%
Fine-tuning	43.48 ± 0.72	86.62 ± 0.21	37.13 ± 0.15	85.58 ± 0.18	34.24 ± 0.45	82.13 ± 0.16	38.28 ± 0.43	84.78 ± 0.18	65.89 $\pm 0.73\%$
VMLM	43.38 ± 0.24	86.65 ± 0.05	38.17 ± 0.15	85.80 ± 0.03	35.69 ± 0.28	82.39 ± 0.05	39.08 ± 0.23	84.95 ± 0.04	64.44 $\pm 0.48\%$
Metric Fine-tuning	42.88 ± 0.55	86.37 ± 0.12	36.63 ± 0.29	85.10 ± 0.29	34.14 ± 0.83	81.88 ± 0.25	37.89 ± 0.55	84.45 ± 0.22	67.89 $\pm 0.29\%$
Word Dropout	43.56 ± 0.26	86.77 ± 0.09	36.77 ± 0.49	85.51 ± 0.16	34.17 ± 0.19	<u>82.23</u> ± 0.11	38.17 ± 0.26	<u>84.83</u> ± 0.09	70.22 $\pm 0.73\%$
Degradation	44.07 ± 0.22	86.71 ± 0.04	37.36 ± 0.11	85.44 ± 0.07	34.18 ± 0.18	82.01 ± 0.04	38.54 ± 0.07	84.72 ± 0.02	67.78 $\pm 0.11\%$
Metric Weighting	43.58 ± 0.61	86.53 ± 0.10	<u>37.42</u> ± 0.15	85.38 ± 0.10	34.12 ± 0.17	81.96 ± 0.18	38.37 ± 0.19	84.62 ± 0.11	<u>69.00</u> $\pm 0.33\%$

Table 1: Results for **English-to-German** in terms of BLEU and COMET on the Multi30k testsets (separately and averaged over the testsets) and the accuracy on the CoMMuTE contrastive dataset. The results are averaged over three separate runs, and $\pm$ shows the standard error. Best results for each base model in each column are marked in **bold**, and the second best are underlined.

• **Source Degradation** - masking specific words in the source sentence during training to enhance the visual sensitivity of the model (Cheng et al., 2024). The method first identifies the words in three categories: color, person, and noun. Second, the identified words are replaced with the mask strings MASK_C, MASK_P, MASK_N for colors, people, and nouns, respectively (in the original paper, the special tokens have been used instead of words). We used the annotations for types of tokens in the Multi30k dataset from Li et al. (2022a) repository⁵.

We first tuned the hyper-parameters of Fine-tuning in the preliminary experiments on the English-to-German language direction and kept them the same for other methods. For Metric-based Fine-tuning, we selected the number of examples for the second-stage fine-tuning in the preliminary experiments and tuned the learning rate separately for each language direction and model out of $\{1e-5, 2e-5, 5e-5, 1e-4\}$. For the VMLM, Word Dropout, and Source Degradation methods, we tuned the probability of masking and dropping out words $p \in \{0.1, 0.2, 0.3\}$. For our method, we first tuned the hyper-parameter $\alpha = 8$ in the preliminary experiments on the English-to-German language direction, and tested three values of the τ hyper-parameter for each language direction and model ($\{5, 6, 7\}$, and $\{1, 1.5, 2\}$ for Qwen3-VL and IDEFICS models, respectively). This is because they depend on the distribution of PCXMI and CPCXMI metrics on the training

github.com/hplt-project/sacre Moses with default parameters.

⁵https://github.com/libeinue/fairseq_mmt.

dataset, which are based on the model’s outputs (see Section 5.1). We determined that the appropriate range for the τ values is between the 80th and 90th percentile of the distribution of maximum metric values in training examples. Further details of the training, including prompts and hyper-parameters can be found in Appendix A.

We measured BLEU⁶ (Papineni et al., 2002) and COMET⁷ (Rei et al., 2020) of the translations on the available Multi30k test sets: *test 2016*, *test 2017*, and *MSCOCO* for English-to-German and English-to-French, and *test 2016* for English-to-Czech (Elliott et al., 2016, 2017; Barrault et al., 2018). We used the default generation parameters for the tested models. To evaluate the visual sensitivity of the fine-tuned models, we employ the CoMMuTE contrastive dataset (Futeral et al., 2023). In this dataset, the models are presented with an image, an ambiguous source sentence, and two possible translations, where the correct one depends on the accompanying image. The dataset measures the accuracy of the models by assigning higher probabilities to correct target sentences. Following Futeral et al. (2023), we repeated the experiments three times with different seeds and averaged the results.

4.1 Results

The results in terms of BLEU and COMET on the *test 2016*, *test 2017*, and *MSCOCO* testsets, and CoMMuTE accuracy are presented in Tables 1 and 2 for the English-to-German and English-

⁶<https://github.com/mjpost/sacrebleu>

⁷We used the Unbabel/wmt22-comet-da model.

Model	test 2016		test 2017		MSCOCO		Average		CoMMuTE Accuracy
	BLEU	COMET	BLEU	COMET	BLEU	COMET	BLEU	COMET	
Qwen3-VL 4B	43.77	86.04	43.47	86.31	43.24	83.69	43.49	85.35	75.00%
Fine-tuning	67.03±0.36	89.81±0.05	61.62±0.13	89.25±0.10	50.90±0.43	85.37±0.12	59.85±0.08	88.14±0.03	81.17±0.37%
VMLM	<u>67.37±0.25</u>	<u>89.71±0.05</u>	61.00±0.19	89.23±0.10	50.71±0.57	85.10±0.07	59.69±0.21	88.01±0.05	81.71±0.39%
Metric Fine-tuning	66.83±0.41	89.43±0.12	60.98±0.11	88.97±0.02	49.73±0.09	84.71±0.15	59.18±0.13	87.70±0.09	81.17±0.19%
Word Dropout	66.46±0.31	89.58±0.09	60.98±0.05	89.10±0.02	<u>51.01±0.44</u>	85.15±0.04	59.48±0.27	87.94±0.04	84.42±0.68%
Degradation	67.42±0.18	89.69±0.04	<u>61.58±0.13</u>	89.26±0.06	51.09±0.18	85.33±0.06	60.03±0.04	88.09±0.02	84.52±0.11%
Metric Weighting	67.22±0.28	89.69±0.03	61.04±0.23	89.15±0.04	50.50±0.37	85.07±0.03	59.59±0.22	87.97±0.03	84.74±0.50%
Qwen3-VL 8B	47.53	86.93	47.32	87.14	46.40	84.36	47.08	86.14	75.00%
Fine-tuning	68.56±0.17	<u>90.06±0.09</u>	62.30±0.15	<u>89.50±0.05</u>	52.66±0.31	85.74±0.17	61.17±0.13	88.43±0.06	83.12±0.00%
VMLM	68.15±0.12	90.01±0.09	61.52±0.08	89.48±0.11	51.56±0.20	85.57±0.08	60.41±0.05	88.35±0.01	81.49±0.19%
Metric Fine-tuning	69.32±0.10	90.07±0.04	62.91±0.27	89.55±0.03	51.31±0.35	<u>85.60±0.10</u>	61.18±0.02	88.41±0.05	84.42±0.50%
Word Dropout	67.20±0.18	89.81±0.05	61.14±0.52	89.03±0.09	51.27±0.15	85.23±0.07	59.87±0.09	88.02±0.04	85.28±0.57%
Degradation	67.29±0.09	89.93±0.04	62.07±0.15	89.29±0.07	50.99±0.18	85.44±0.09	60.12±0.02	88.22±0.06	81.28±0.29%
Metric Weighting	67.74±0.16	89.98±0.06	<u>62.68±0.19</u>	89.37±0.03	51.19±0.18	85.56±0.16	60.54±0.07	88.30±0.04	<u>85.06±0.50%</u>
IDEFICS 9B	33.35	82.26	32.15	81.76	31.80	78.64	32.43	80.89	71.75%
Fine-tuning	67.00±0.19	<u>89.39±0.11</u>	60.39±0.13	<u>88.62±0.09</u>	<u>50.58±0.24</u>	84.87±0.09	59.32±0.12	87.63±0.03	76.73±0.22%
VMLM	66.78±0.04	89.47±0.05	60.44±0.05	88.68±0.11	50.83±0.36	85.03±0.15	59.35±0.12	87.73±0.02	78.57±0.37%
Metric Fine-tuning	67.52±0.24	89.47±0.16	60.19±0.18	88.54±0.02	50.41±0.38	84.94±0.16	59.37±0.25	87.65±0.11	77.92±0.32%
Word Dropout	65.82±0.12	89.19±0.06	59.74±0.11	88.43±0.03	49.82±0.24	84.80±0.19	58.46±0.08	87.47±0.05	78.68±0.43%
Degradation	67.00±0.13	89.28±0.03	60.50±0.07	88.54±0.01	50.08±0.33	84.96±0.13	59.19±0.09	87.59±0.05	78.14±0.89%
Metric Weighting	66.55±0.12	89.22±0.05	60.29±0.07	88.48±0.08	50.51±0.22	85.35±0.08	59.12±0.05	<u>87.69±0.05</u>	78.79±0.29%

Table 2: Results for **English-to-French** in terms of BLEU and COMET on the Multi30k testsets (separately and averaged over the testsets) and the accuracy on the CoMMuTE contrastive dataset. The results are averaged over three separate runs, and $\pm$ shows the standard error. Best results for each base model in each column are marked in **bold**, and the second best are underlined.

Model	test 2016		CoMMuTE Accuracy
	BLEU	COMET	
Qwen3-VL 4B	23.12	83.43	67.21%
Fine-tuning	35.82±0.25	88.63±0.05	74.35±0.37%
VMLM	35.66±0.08	88.46±0.10	74.68±0.50%
Metric Fine-tuning	35.19±0.16	88.39±0.06	<u>75.54±1.07%</u>
Word Dropout	35.72±0.20	88.62±0.09	75.43±0.66%
Degradation	36.19±0.40	<u>88.62±0.02</u>	74.89±0.29%
Metric Weighting	36.19±0.31	88.46±0.12	76.41±0.43%
Qwen3-VL 8B	26.82	86.37	74.03%
Fine-tuning	36.17±0.08	89.06±0.06	78.03±0.22%
VMLM	36.99±0.34	88.84±0.13	78.46±0.29%
Metric Fine-tuning	36.42±0.15	88.64±0.03	79.98±0.22%
Word Dropout	35.98±0.29	88.91±0.12	80.84±0.19%
Degradation	36.68±0.52	88.87±0.12	78.57±0.19%
Metric Weighting	36.52±0.24	88.81±0.03	<u>80.19±0.32%</u>
IDEFICS 9B	10.29	70.42	64.29%
Fine-tuning	36.80±0.18	88.27±0.04	68.72±0.76%
VMLM	36.41±0.16	88.35±0.11	70.02±0.47%
Metric Fine-tuning	36.33±0.33	88.16±0.09	71.43±0.32%
Word Dropout	36.89±0.06	88.65±0.09	<u>73.27±0.66%</u>
Degradation	36.66±0.05	88.28±0.09	72.84±0.92%
Metric Weighting	36.76±0.18	88.15±0.08	74.03±0.32%

Table 3: Results for **English-to-Czech** in terms of BLEU and COMET on the Multi30k test 2016 testset and the accuracy on the CoMMuTE contrastive dataset. The results are averaged over three separate runs, and $\pm$ shows the standard error. Best results for each base model in each column are marked in **bold**, and the second best are underlined.

to-French language directions. To improve the readability of the results, we additionally included the BLEU and COMET averaged over the testsets. The results in terms of BLEU and COMET on the *test 2016* testset and CoMMuTE accuracy for the English-to-Czech language direction are shown in Table 3. Results are averaged over three runs for each method, and standard errors are reported in the tables.

In terms of BLEU and COMET, Qwen3-VL 8B exhibited better performance than the other two models, with IDEFICS 9B scoring lower than the smaller Qwen-VL 4B model on all language di-

rections. While this could be caused by different generation method (no sampling for the IDEFICS model), this gap is removed after fine-tuning on the Multi30k dataset. The CoMMuTE accuracy of the baseline models is similar between English-to-German and English-to-French, and there is a clear drop for English-to-Czech for Qwen-3-VL 4B and IDEFICS 9B models (more than 6 and 4 percentage points, respectively). This could be attributed to lower amounts of English-to-Czech parallel and Czech monolingual data in the training corpora.

For English-to-German, fine-tuning reduces the visual grounding, as measured by the CoMMuTE accuracy, compared to the baseline model. Only Word Dropout and Metric-based Loss Weighting retrieve and surpass the performance of the baseline model. This effect is not visible for the two remaining language directions, where each technique improves the baseline performance. We hypothesize that the drop in CoMMuTE accuracy from English to German stems from linguistic proximity (both are Germanic languages). This proximity may make the translation task easier for the models to learn during fine-tuning, reducing their reliance on the visual modality to resolve ambiguity. We leave evaluating this hypothesis for future work. VMLM did not yield large improvements in visual sensitivity compared to other methods. We attribute this to our use of only a parallel multimodal dataset for this task. We expect it to show larger improvements when trained on a large-scale monolingual multimodal dataset. However, training it already used twice as much computational resources compared to other methods (see Appendix A), and scal-

ing it further falls outside the scope of this work.

While the highest results of CoMMuTE accuracy are significantly different from the baseline models and Fine-tuning (according to Welch’s t-test), due to the low number of runs, the results for the top methods (marked as bold and underlined in the result tables) are not significantly different. Nevertheless, Metric-based Loss Weighting achieved the highest results in terms of CoMMuTE accuracy on 6 out of 9 experimental settings, and achieved the second-best result for the remaining three. Word Dropout showed the best result in three settings and was the second best in four of them. Source Degradation achieved the second best result on English-to-French with the Qwen3-VL 4B model, and Metric-based Fine-tuning on English-to-Czech with the same model. Notably, the high score on CoMMuTE does not correlate well with the achieved BLEU and COMET scores, indicating that the general translation quality and visual grounding are not necessarily mutual, which corroborates the findings of [Futeral et al. \(2025\)](#). This can be attributed to the examples in the Multi30k testsets not requiring images for good translation ([Caglayan et al., 2019](#); [Wu et al., 2021](#)). We further investigate this phenomenon in Section 5.2, where we qualitatively analyze the selected translations in the MSCOCO English-to-French dataset.

5 Analysis

In this section, we take a deeper look at the PCXMI and CPCXMI distributions. Next, we examine translation examples produced by our method and standard fine-tuning. Lastly, we experiment with combining our method with Word Dropout, another technique that achieves strong results. Additionally, in Appendix B we present the ablation study.

5.1 Metrics Analysis

In Figure 2, we present two examples of the per-token values of the PCXMI and CPCXMI metrics. Further examples can be found in Appendix C. While both metrics are capable of identifying the tokens that require image for disambiguation, they are not perfect, and can ignore or misidentify those tokens. This can be partially mitigated by taking the maximum of both metrics.

In Appendix C, we present the distributions of the calculated metrics (PCXMI, CPCXMI, and their maximum) on the training subset of the

Multi30k dataset for each tested model and language direction. The values of the metrics depend heavily on the model, as can be seen in the difference between both the Qwen3-VL models and IDEFICS 9B. The difference between language directions for the same model is a lot smaller. Additionally, all distributions show a concentration of values close to zero and a long tail at high values. This means that most examples in the dataset do not require, according to the tested models, images to disambiguate. The distributions for Multi30k and CoMMute datasets are compared in Appendix C, where we show that the distribution for the latter is denser for higher values of the metric.

5.2 Qualitative Analysis

In this section, we investigate the drop in BLEU exhibited by Metric-based Loss Weighting compared to Fine-tuning seen for Qwen3-VL 8B on the English-to-French language direction. We perform the qualitative analysis of the translations of the MSCOCO testset, where the difference in BLEU between the methods is the largest. In Figure 3, we present the distributions of per-example BLEU scores for Fine-tuning and our method. They differ mostly in two regions: (1) BLEU scores around 100, where Fine-tuning has more examples, (2) BLEU scores around 30, where the Weighting distribution is denser.

We examined the examples that exhibit the highest difference in BLEU between Fine-tuning and our method (both ways, with both Fine-tuning and our method achieving better scores). We found that the largest drop in example-level BLEU scores mostly corresponds to acceptable translations that differ from the gold target translation. More examples with our descriptions and further details of the analysis are presented in Appendix D.

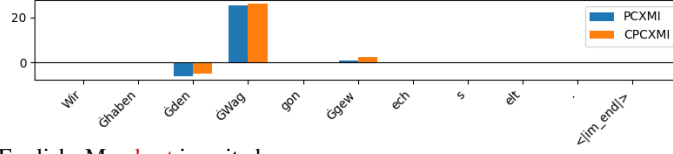
5.3 Combining Weighting with Dropout

The main results of the tested methods show that Metric-based Loss Weighting achieves the highest CoMMuTE accuracy. The second best performing method, Word Dropout, modifies the source side and is technically not related to our method, which changes the training signal corresponding to the target side. In this section, we test whether the combination of both methods could increase the visual sensitivity of the resulting models even further.

For each example, the Metric-based Loss Weighting is applied (as described in Section 3).



English: We changed **coaches**.
German: Wir haben den **Waggon** gewechselt.



English: My **chest** is quite heavy...
German: Mein **Koffer** ist ziemlich schwer.

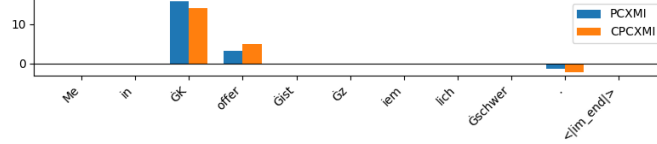


Figure 2: Two examples from the English-to-German CoMMuTE dataset with the per-token values of PCXMI and CPCXMI metrics for the Qwen3-VL 4B model. In red we mark the ambiguous words in the source side and the corresponding words in the target side.

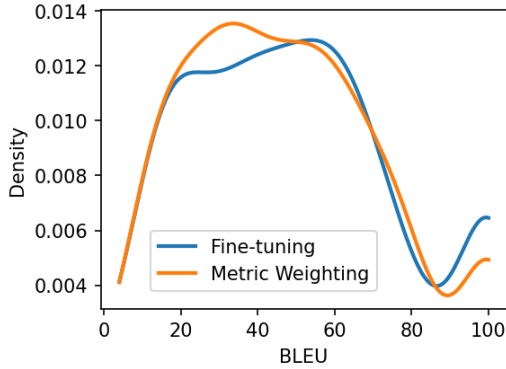


Figure 3: Distributions of BLEU scores of Fine-tuning and Metric-based Loss Weighting on the MSCOCO testset for the English-to-French language direction.

In cases where there is no token with the metric value greater than the threshold parameter τ , we apply Word Dropout to the source sentence with the specified per-token probability p . We test the combined method with the Qwen3-VL 4B model on all three tested language directions (the experimental configuration is the same as before). The results can be seen in Table 4, where we compare with Fine-tuning, and both methods (the full non-aggregated results for English-to-German and English-to-French can be found in Appendix E). The combined method achieves higher CoMMuTE accuracy on English-to-German and English-to-Czech, but did not improve the Metric-based Loss Weighting on English-to-French. Furthermore, the combined method shows reduced BLEU for all tested languages, but the reduction in COMET was not observed. This shows that both methods can work in tandem, increasing the visual grounding even further, but resulting in the distributions of

outputs moving away from the gold translations.

Model	Average		CoMMuTE Accuracy
	BLEU	COMET	
English-to-German			
Qwen3-VL 4B	33.80	83.90	74.00%
Fine-tuning	38.64 ±0.18	85.11±0.05	71.67±0.19%
Word Dropout	38.13±0.18	85.03±0.04	76.89±0.44%
Metric Weighting	38.30±0.15	85.34 ±0.07	77.00±0.33%
Weighting + Dropout	37.80±0.11	85.23 ±0.09	78.78 ±0.11%
English-to-French			
Qwen3-VL 4B	43.49	85.35	75.00%
Fine-tuning	59.85 ±0.08	88.14 ±0.03	81.17±0.37%
Word Dropout	59.48±0.27	87.94±0.04	84.42±0.68%
Metric Weighting	59.59±0.22	87.97±0.03	84.74 ±0.50%
Weighting + Dropout	58.71±0.22	87.93±0.09	84.63±0.39%
English-to-Czech			
Qwen3-VL 4B	23.12	83.43	67.21%
Fine-tuning	35.82±0.25	88.63±0.05	74.35±0.37%
Word Dropout	35.72±0.20	88.62±0.09	75.43±0.66%
Metric Weighting	36.19 ±0.31	88.46±0.12	76.41 ±0.43%
Weighting + Dropout	35.27±0.39	88.69 ±0.11	77.27 ±0.37%

Table 4: Results for combined Metric-based Loss Weighting and Word Dropout in terms of BLEU and COMET averaged over the test sets (and three separate runs) for English-to-German and English-to-French, and for *test 2016* for English-to-Czech, and CoMMuTE accuracy.

6 Conclusions

This work introduced a training strategy for fine-tuning MLLMs for the task of MMT that increases the visual grounding during translation. It increases the loss for the tokens that benefit from context in the form of the image, as measured by the Point-wise Cross-Mutual Information (PCXMI) and Congruency-based PCXMI (CPCXMI). Both metrics rely on the model’s output probabilities with and without the image (or with incongruent image in case of CPCXMI).

We empirically tested our technique against existing methods adapted for the task of fine-tuning MLLMs on three models and three language directions, and showed its increased performance

in terms of CoMMuTE accuracy, while maintaining strong overall translation quality, as measured by BLEU and COMET. Our results confirm that general-purpose translation metrics (BLEU and COMET) fail to capture how strongly models rely on visual modality for MMT. In future work, we will explore how the metrics can inform masking and word dropout on the source side.

7 Limitations

While we investigate three language directions and three sub-families, all of them come from the Indo-European family, and all use the Latin script. This limitation was imposed by the language directions covered by both Multi30k and CoMMuTE datasets. We suspect that the measured performance of the methods goes beyond the tested language pairs, but this claim has not been tested experimentally. Due to the low number of repeated runs and the variability of the measured metrics, the results of the top methods are not different with statistical significance. We limited our investigation to only the same multimodal parallel dataset for each method, while VMLM could use additional multimodal monolingual data.

For Metric-based Loss Weighting, we calculate the image dependence metrics using the baseline models. Updating the metrics during training or using a separate (presumably more visually sensitive) model could yield different results, but we did not test this setting. Furthermore, we limit our method to PCXMI and CPCXMI, and a broader exploration of image dependence metrics could lead to increased performance. Finally, while we tested the methods on three MLLMs, their sizes were limited to less than 10B parameters.

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A Training Details

We trained the models using the AdamW optimizer (Loshchilov and Hutter, 2019). We used a single NVIDIA GeForce RTX 3090 24GB for Qwen3-VL 4B, and a single NVIDIA A100 40GB for Qwen3-VL 8B and IDEFICS 9B. The average training times per epoch for each model are presented in Table 5. The time for Metric-based Fine-tuning is shown only for the second stage training (training on the top-k examples). The training time for VMLM is roughly doubled compared to fine-tuning, because of the auxiliary task of masked

language modeling. The times for other methods are comparable, meaning that additional pre-processing (dropout, masking) and post-processing (loss weighting) do not meaningfully increase the training time compared to basic fine-tuning. The training was performed on the Multi30k dataset and evaluated on its testsets and the CoMMuTE contrastive dataset. The sizes of the used datasets can be found in Table 6.

Model	Epoch Time
Qwen3-VL 4B	
Fine-tuning	1:43
VMLM	3:30
Metric Fine-tuning*	0:20
Word Dropout	1:43
Degradation	1:43
Metric Weighting	1:43
Qwen3-VL 8B	
Fine-tuning	1:16
VMLM	2:24
Metric Fine-tuning*	0:13
Word Dropout	1:14
Degradation	1:15
Metric Weighting	1:14
IDEFICS 9B	
Fine-tuning	0:51
VMLM	1:40
Metric Fine-tuning*	0:09
Word Dropout	0:49
Degradation	0:50
Metric Weighting	0:51

Table 5: Average training times per epoch for the tested models (in hours:minutes). *The time for Metric-based Fine-tuning is only for the second stage of the training.

Dataset	# Examples
Multi30k	
Train	29,000
Validation	1014
Test 2016	1000
Test 2017	1000
MSCOCO	461
CoMMuTE	
Test	300
Validation	60

Table 6: The sizes of the used datasets in terms the number of examples.

The shared hyper-parameters for all methods and

models can be found in Table 7. For generation, we used the default parameters of all models: multinomial sampling with temperature 0.7 for Qwen3-VL models, and greedy decoding for IDEFICS 9B. The translation prompt can be seen in Listing 1, and the VMLM prompt in Listing 2.

Listing 1: The translation prompt used for generation, contrastive evaluation on CoMMuTE, and metric calculation. The source and target languages (in natural language, e.g., "English" or "German") are represented as {src_lang} and {tgt_lang}, respectively. The source sentence is represented as {src}. The accompanying image is passed as the input alongside the prompt.

```
"Translate the following {src_lang} text
  → to {tgt_lang}.\n
{src_lang}: {src}\n
{tgt_lang}:"
```

Listing 2: The Visually Conditioned Masked Language Modeling (VMLM) prompt and response used for training. The source language (in natural language, e.g., "English") and masked source text are represented as {src_lang} and {masked_text}, respectively. {masked_word1} and {masked_word2} represent the masked words (the number of masked words can be different between examples). The accompanying image is passed as the input alongside the prompt.

```
Prompt:
"The following {src_lang} sentence has
  → some words replaced with mask
  → tokens.\n
Recover the masked words.\n
Sentence: {masked_text}"

Response:
"MASK1: {masked_word1}, MASK2:{
  → masked_word2}, ..."
```

We tuned the specific hyper-parameters for each method:

- **Visually Conditioned Masked Language Modeling (VMLM)** - token masking probability $p \in \{0.1, 0.2, 0.3\}$;
- **Metric-based Fine-tuning** - learning rate values $\{1e-5, 2e-5, 5e-5, 1e-4\}$ with the constant number of top-k examples for the second stage fine-tuning $k = 5000$;
- **Word Dropout** - token dropout probability $p \in \{0.1, 0.2, 0.3\}$;
- **Source Degradation** - token masking probabil-

ity $p \in \{0.1, 0.2, 0.3\}$ for each eligible word;

- **Metric-based Loss Weighting** - threshold $\tau \in \{5, 6, 7\}$ for Qwen3-VL models and $\tau \in \{1, 1.5, 2\}$ for IDEFICS 9B (based on the PCXMI and CPCXMI distributions) with constant loss weighting $\alpha = 8$.

The tuning was performed based on the validation subsets of the Multi30k and CoMMuTE datasets. First, we took only the runs (averaged over 3 seeds) with the highest BLEU on the Multi30k validation (the values not lower than 0.5 BLEU below the highest for a particular model and language direction). Next, we selected the run with the highest accuracy on the CoMMuTE validation subset. While the validation dataset for Multi30k is available for all tested language directions, CoMMuTE only provides it for English-to-French. For this reason, we translated the target side (French) to both German and Czech using Gemini 3.5 Flash⁸ through Google API. The prompt for translation is presented in Listing 3. The found best hyper-parameters for each method and language direction can be found in Table 8.

Listing 3: The prompt used to translate the CoMMuTE validation set from French to German and Czech. The source and target expressions in the original dataset are represented as {src} and {tgt}, respectively. The translation language (German and Czech) is marked as {trans_lang}. The accompanying image is passed as the input alongside the prompt.

```
"Translate the following English
  ↪ expression to {trans_lang}: {src}
  ↪ }. \n
Use the image and the following French
  ↪ translation as references: {tgt}
  ↪ }. \n
Output only the translation."
```

B Ablation Study

In Figure 4, we present the results of the ablation study of the Metric-based Loss Weighting method in terms of BLEU and COMET scores averaged over all testsets, and CoMMuTE accuracy. We trained Qwen3-VL 4B model using our method with different hyper-parameters: type of metric (PCXMI, CPCXMI, and their maximum), $\alpha \in \{2, 4, 8, 16\}$, and $\tau \in \{4, 5, 6, 7\}$. The results show that using the maximum of both metrics

⁸gemini-3.5-flash accessed on 10.07.2026

Hyper-parameter	Value
Optimizer	AdamW
Learning Rate	0.0001*
LR Scheduler	Cosine
LR Warmup Steps	500
Weight Decay	0.01
Max Grad Norm	1.0
Batch Size	8
Gradient Accumulation Steps	8
Num Epoch	10
Patience	2
Precision	bf16
Lora α	64
Lora r	32
Lora Dropout	0.05
Max Length	200
Seeds	1, 2, 3

Table 7: The shared hyper-parameters of all tested fine-tuning methods. *We tuned the Learning Rate hyper-parameter for the second stage of the Metric-based Fine-tuning method.

yield the higher CoMMuTE accuracy than using the metrics separately. The results for the α and τ hyper-parameters are stable, but we make the following observations. Increasing the α parameter lead to the increase in the CoMMuTE accuracy but can reduce the BLEU and COMET scores when set too high. High τ values result in high BLEU and COMET scores, but lower CoMMuTE accuracies. This is expected as τ acts as a threshold for the token selection, and high values correspond to the lower number of tokens with increased loss.

C Extended Metrics Analysis

In Figure 7, we present more examples of per-token values of the metrics from the CoMMuTE dataset. The metrics are capable of correctly identifying the image-dependent tokens in the target sentences.

Figure 5 shows the distribution of values of the metrics (PCXMI, CPCXMI and the maximum of both) on the Multi30k training dataset. We take the maximum value of a metric of the target tokens for each example in the dataset. For both Qwen3-VL models and all language directions, the CPCXMI is lower than PCXMI, showing higher density for values close to zero. This is not the case for IDEFICS 9B, where the distributions are very similar for all language directions.

The comparison between the distributions of the maximum of PCXMI and CPCXMI on the training

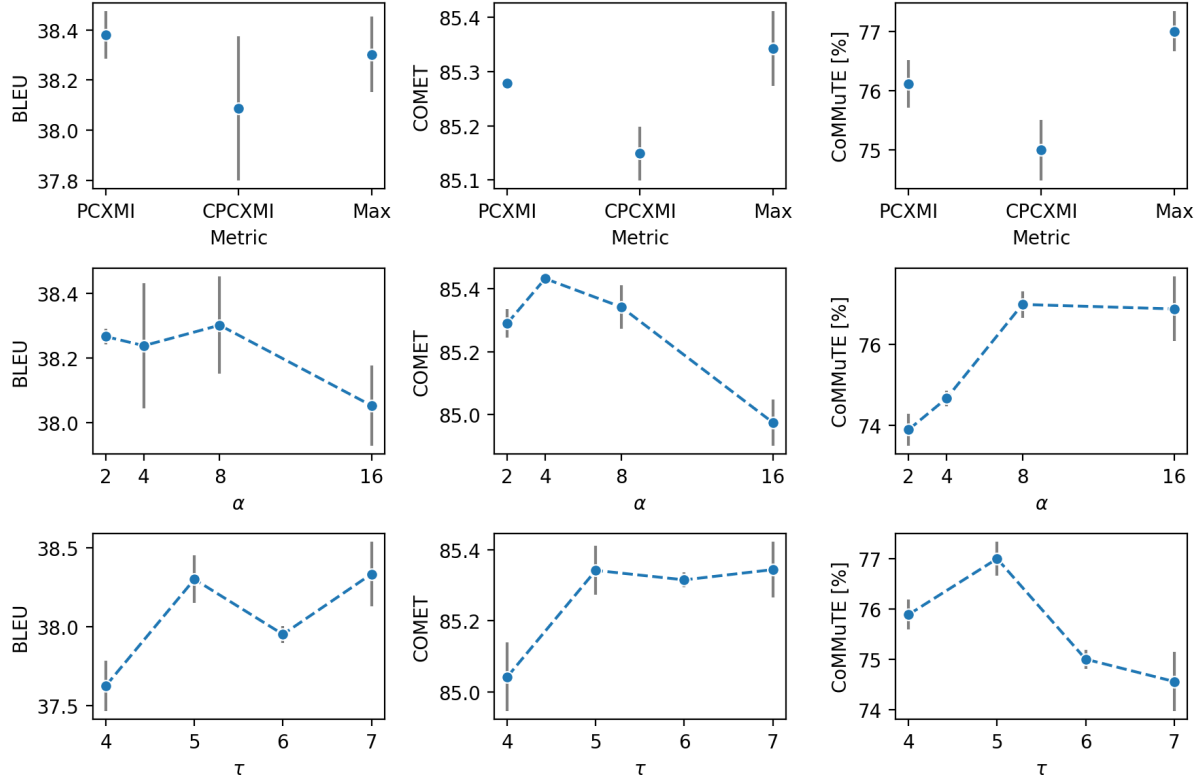


Figure 4: The results of applying the Metric-based Loss Weighting method to Qwen3-VL 4B model on the English-to-German language direction with different values of hyper-parameters. The results are in terms of the BLEU and COMET scores averaged over all used testsets, and the accuracy on CoMMuTE dataset. We averaged the results over three separate runs and error bars show the standard error.

Model	En-De	En-Fr	En-Cs
Qwen3-VL 4B			
VMLM (p)	0.3	0.1	0.1
Metric Fine-tuning (lr)	1e-4	1e-4	1e-5
Word Dropout (p)	0.1	0.1	0.1
Degradation (p)	0.3	0.2	0.1
Metric Weighting (τ)	5	6	6
Qwen3-VL 8B			
VMLM (p)	0.1	0.1	0.3
Metric Fine-tuning (lr)	1e-5	2e-5	5e-5
Word Dropout (p)	0.1	0.1	0.1
Degradation (p)	0.3	0.3	0.2
Metric Weighting (τ)	5	6	5
IDEFICS 9B			
VMLM (p)	0.1	0.2	0.3
Metric Fine-tuning (lr)	1e-5	2e-5	2e-5
Word Dropout (p)	0.1	0.1	0.1
Degradation (p)	0.2	0.2	0.3
Metric Weighting (τ)	1.5	2	1.5

Table 8: Found best hyper-parameters for each method and language direction. For VMLM, Word Dropout and Degradation, the hyper-parameter is the probability of masking or dropout p. For Metric-based Fine-tuning, the hyper-parameter is the learning rate (lr) of the second stage training. For Metric-based Weighting, the hyper-parameter is the threshold τ .

split of Multi30k and the CoMMuTE dataset is shown in Figure 6. The metric for CoMMuTE is not as concentrated close to zero as for Multi30k, which is consistent with the presence of ambiguous words in every example of the CoMMuTE dataset.

D Extended Qualitative Analysis

To examine the source of the largest measured drop in BLEU between Fine-tuning and Metric-based Loss Weighting in our experiments, we analyzed the translations of Qwen3-VL 8B on *MSCOCO* English-to-French testset of the runs with seed 1, which also shows the highest BLEU difference. We looked at the 20 examples with the highest difference of example-level BLEU scores between the two methods (20 in each direction). The selected instances for Fine-tuning scoring higher can be seen in Figure 8, and in Figure 9 for Weighting scoring higher. We categorize the causes of the drop in BLEU into three categories: (1) change of the tense (mostly present continuous instead of present simple), (2) synonyms or equivalent expressions, (3) true errors including omissions and incorrect

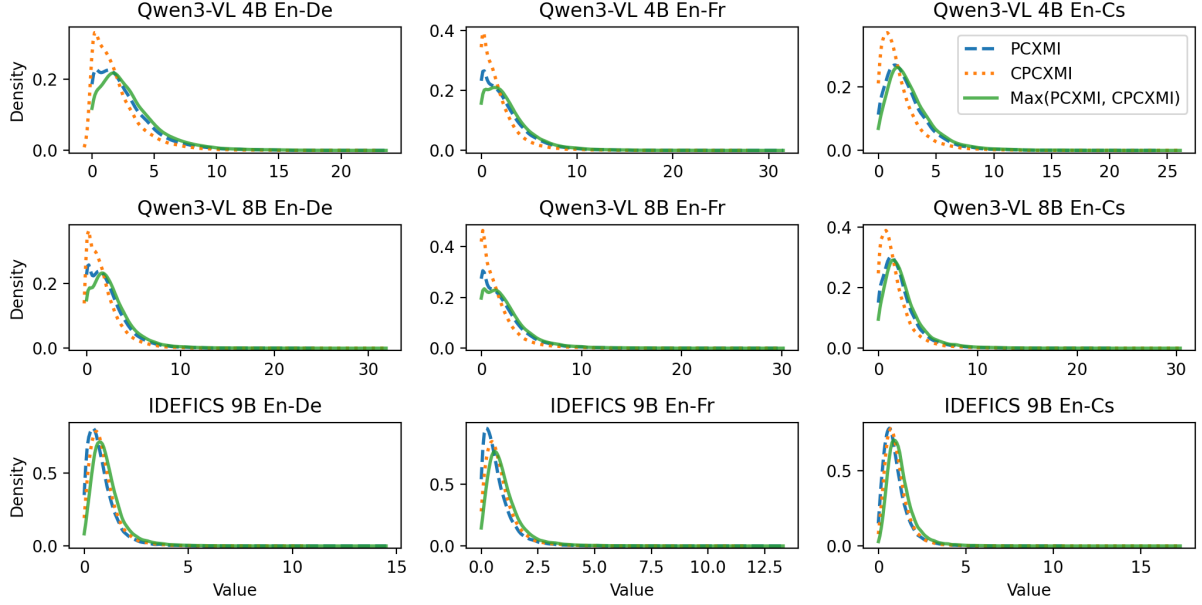


Figure 5: Kernel Density Estimate visualization of distributions of **metrics (PCXMI, CPCXMI, and the maximum of both)** in the Multi30k training dataset for each tested model and language direction. We take the maximum token-level metric for each example in the dataset.

meaning. Out of the 20 instances with highest drop of BLEU for Weighting 30% were due to the use of a different form, 45% were equivalent, and 25% were erroneous. From the 20 instances where Fine-tuning showed the largest drop in BLEU 10% were due to a different form, 50% were synonyms, and 40% contained errors. While this analysis only looked at the selected examples, it shows that the largest drop in example-level BLEU scores mostly corresponds to acceptable translations that differ from the gold target translation.

E Extended Results

In this section, we present the full results for the combined method (Metric-based Loss Weighting and Word Dropout). Table 9 shows the results for all testsets on the English-to-German and English-to-French language directions for the Qwen3-VL 4B model.

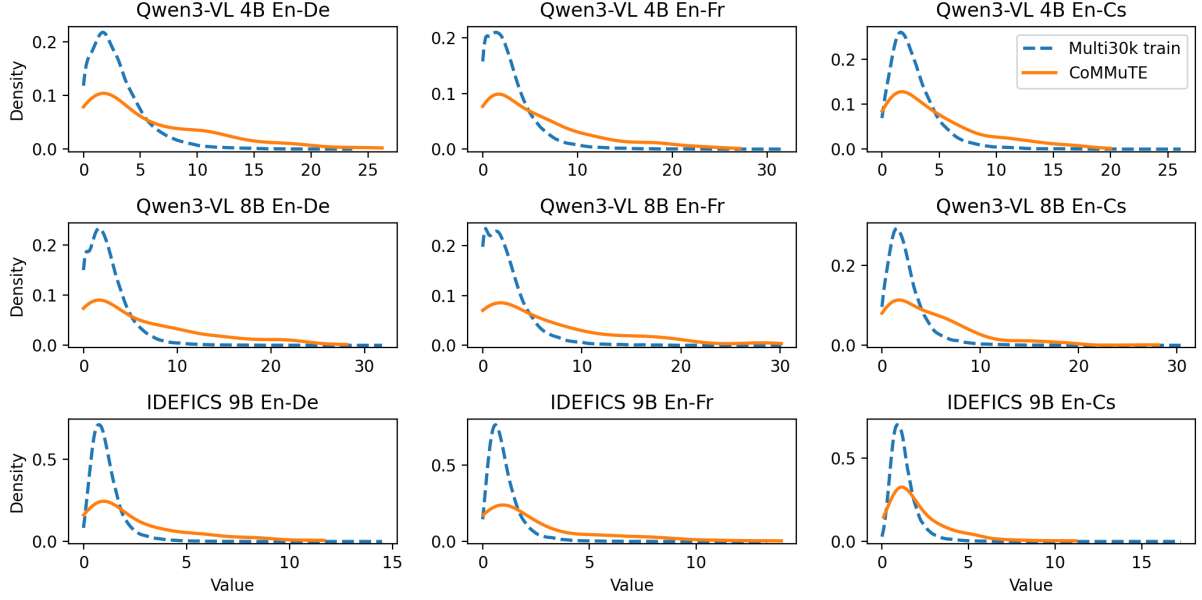
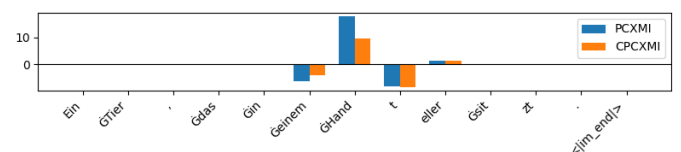
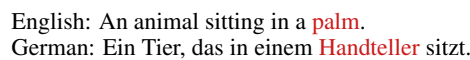
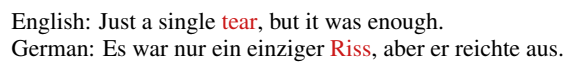
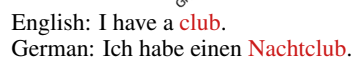
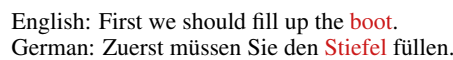
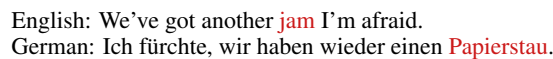
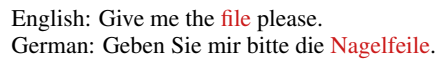
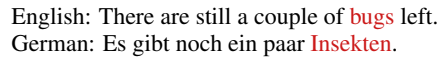


Figure 6: Kernel Density Estimate visualization of distributions of Maximum of PCXMI and CPCXMI in the **Multi30k training and CoMMuTE** datasets for each tested model and language direction. We take the maximum token-level metric for each example in the dataset.

Model	test 2016		test 2017		MSCOCO		Average		CoMMuTE Accuracy
	BLEU	COMET	BLEU	COMET	BLEU	COMET	BLEU	COMET	
English-to-German									
Qwen3-VL 4B	38.51	85.56	32.95	84.90	29.94	81.24	33.80	83.90	74.00%
Fine-tuning	42.61 ±0.12	86.66±0.06	38.08 ±0.38	85.98±0.06	<u>35.22</u> ±0.45	82.70±0.13	38.64 ±0.18	85.11±0.05	71.67±0.19%
Word Dropout	<u>42.20</u> ±0.19	86.62±0.06	37.17±0.19	85.80±0.01	35.00±0.22	82.69±0.16	38.13±0.18	85.03±0.04	76.89±0.44%
Metric Weighting	41.91±0.33	<u>86.80</u> ±0.04	<u>37.29</u> ±0.14	86.14 ±0.04	35.71 ±0.19	83.09 ±0.13	<u>38.30</u> ±0.15	85.34 ±0.07	<u>77.00</u> ±0.33%
Weighting + Dropout	41.56±0.49	86.87 ±0.07	36.85±0.36	<u>86.04</u> ±0.07	34.99±0.06	<u>82.80</u> ±0.19	37.80±0.11	<u>85.23</u> ±0.09	78.78 ±0.11%
English-to-French									
Qwen3-VL 4B	43.77	86.04	43.47	86.31	43.24	83.69	43.49	85.35	75.00%
Fine-tuning	<u>67.03</u> ±0.36	89.81 ±0.05	61.62 ±0.13	89.25 ±0.10	<u>50.90</u> ±0.43	85.37 ±0.12	59.85 ±0.08	88.14 ±0.03	81.17±0.37%
Word Dropout	66.46±0.31	89.58±0.09	60.98±0.05	89.10±0.02	51.01 ±0.44	85.15±0.04	59.48±0.27	87.94±0.04	84.42±0.68%
Metric Weighting	67.22 ±0.28	89.69±0.03	<u>61.04</u> ±0.23	<u>89.15</u> ±0.04	50.50±0.37	85.07±0.03	<u>59.59</u> ±0.22	<u>87.97</u> ±0.03	84.74 ±0.50%
Weighting + Dropout	65.74±0.39	89.49±0.10	59.69±0.07	89.12±0.07	50.71±0.32	<u>85.19</u> ±0.19	58.71±0.22	87.93±0.09	<u>84.63</u> ±0.39%

Table 9: Results for combined Metric-based Loss Weighting and Word Dropout for all testsets in English-to-German and English-to-French. The results are averaged over three separate runs.



18



English: A young woman on a snowboard putting on gloves.
 French: Une jeune femme sur un snowboard mettant des gants.
 Fine-tuning: Une jeune femme sur un snowboard mettant des gants.
 Weighting: Une jeune femme **en** snowboard **en train de** mettre des gants.
 BLEU: 100.00/22.03 (-77.97) | COMET: 96.36/86.10 (-10.26)
 Weighting used present continuous.



English: A person that is playing in a tennis game.
 French: Une personne qui joue un match de tennis.
 Fine-tuning: Une personne qui joue un match de tennis.
 Weighting: Une personne qui joue au tennis.
 BLEU: 100.00/36.74 (-63.26) | COMET: 97.80/81.47 (-16.33)
 Weighting did not include the tennis match.



English: A man pulling a pizza out of an oven.
 French: Un homme sortant une pizza d'un four.
 Fine-tuning: Un homme sortant une pizza d'un four.
 Weighting: Un homme **en train de** sortir une pizza d'un four.
 BLEU: 100.00/40.35 (-59.65) | COMET: 97.05/92.57 (-4.48)
 Weighting used present continuous.



English: There is a man putting on a tie.
 French: Il y a un homme qui met une cravate.
 Fine-tuning: Il y a un homme qui met une cravate.
 Weighting: Il y a un homme **en train de** mettre une cravate.
 BLEU: 100.00/42.40 (-57.60) | COMET: 95.79/79.28 (-16.51)
 Weighting used present continuous.



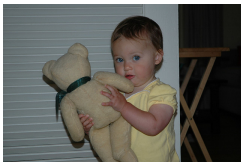
English: A dog being blown dry by a man.
 French: Un chien séché par un homme.
 Fine-tuning: Un chien séché par un homme.
 Weighting: Un chien **en train d'être** séché par un homme.
 BLEU: 100.00/45.18 (-54.82) | COMET: 95.76/75.62 (-20.14)
 Weighting used present continuous.



English: A table set for a dinner of pizza.
 French: Une table dressée pour un dîner de pizza.
 Fine-tuning: Une table dressée pour un dîner de pizza.
 Weighting: Une table dressée pour un dîner **avec** des pizzas.
 BLEU: 100.00/53.73 (-46.27) | COMET: 97.67/92.51 (-5.16)
 Weighting translated as "dinner with pizzas".



English: A man is reading a book while another is holding a dog.
 French: Un homme est en train de lire un livre pendant qu'un autre tient un chien.
 Fine-tuning: Un homme est en train de lire un livre tandis qu'un autre tient un chien.
 Weighting: Un homme lit un livre tandis qu'un autre tient un chien.
 BLEU: 81.54/36.95 (-44.59) | COMET: 92.98/86.83 (-6.15)
 Weighting used present simple.



English: Baby in yellow holds on to her teddy bear.
 French: Le bébé en jaune tient son ours en peluche.
 Fine-tuning: Un bébé en jaune tient son ours en peluche.
 Weighting: Un bébé en jaune **se** tient **à** son ours en peluche.
 BLEU: 88.01/43.67 (-44.34) | COMET: 92.68/85.95 (-6.73)
 Weighting changed the meaning to "supporting oneself".



English: A man covering his eyes next to cases of bananas
 French: Un homme couvrant ses yeux à côté des boîtiers de bananes
 Fine-tuning: Un homme couvrant ses yeux à côté de caisses de bananes
 Weighting: Un homme **se** couvrant **les** yeux à côté de caisses de bananes
 BLEU: 63.16/19.16 (-44.00) | COMET: 89.27/85.48 (-3.79)
 Weighting used the reflexive form.

Figure 8: Selected examples with the highest difference in BLEU between Fine-tuning and Metric-based Loss Weighting (Fine-tuning scoring higher) in the MSCOCO testset on the English-to-French language direction of Qwen3-VL 8B. We also report BLEU and COMET scores for the translations by the methods (and their differences), and our description of errors.



English: A pizza sitting next to a bowl of salad.
 French: Une pizza posée à côté d'un bol de salade.
 Fine-tuning: Une pizza placée à côté d'une salade.
 Weighting: Une pizza posée à côté d'un bol de salade.
 BLEU: 14.90/100.00 (85.10) | COMET: 88.46/97.40 (8.94)
 Fine-tuning ommited the bowl (of salad)



English: A cop leads the way for other people on motorcycles.
 French: Un policier ouvre la voie à d'autres personnes sur des motos.
 Fine-tuning: Un policier **dirige** la **route** pour d'autres personnes en motos.
 Weighting: Un policier ouvre la voie pour d'autres personnes sur des motos.
 BLEU: 12.32/73.49 (61.17) | COMET: 85.39/93.69 (8.30)
 Fine-tuning used the incorrect verb "to direct".



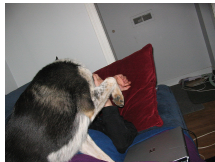
English: a man moving a refrigerator in to a house
 French: un homme transportant un réfrigérateur dans d'une maison
 Fine-tuning: Un homme **en train de** mettre un réfrigérateur dans une maison
 Weighting: un homme transportant un réfrigérateur dans une maison
 BLEU: 15.85/70.71 (54.86) | COMET: 78.21/92.69 (14.48)
 Fine-tuning used present continuous and a synonym.



English: A black cat climbing into a bath tub.
 French: Un chat noir grimpant dans une baignoire.
 Fine-tuning: Un chat noir **montant** dans une baignoire.
 Weighting: Un chat noir grimpant dans une baignoire.
 BLEU: 50.00/100.00 (50.00) | COMET: 86.99/96.85 (9.86)
 Fine-tuning used a synonym.



English: A camera set up in front of a sleeping cat.
 French: Un appareil photo installé devant un chat qui dort.
 Fine-tuning: **Une caméra** installée devant un chat endormi.
 Weighting: Un appareil photo installé devant un chat endormi.
 BLEU: 17.20/67.17 (49.97) | COMET: 72.14/73.10 (0.96)
 Fine-tuning used incorrect "video camera".



English: A dog climbing on top of a person sleeping on a couch.
 French: Un chien qui grimpe sur une personne qui dort sur un canapé.
 Fine-tuning: Un chien **escaladant** un **individu** dormant sur un canapé.
 Weighting: Un chien grimpant sur une personne qui dort sur un canapé.
 BLEU: 23.14/70.77 (47.62) | COMET: 71.39/90.27 (18.88)
 Fine-tuning used incorrect in the context "climbing (e.g., on a mountain)" and a synonym.



English: A blue, yellow and green surfboard sticking out of a sandy beach.
 French: Une planche de surf bleue, jaune et verte plantée dans le sable d'une plage.
 Fine-tuning: Une planche de surf bleue, jaune et verte **dépassant** d'une plage de sable.
 Weighting: Une planche de surf bleue, jaune et verte plantée dans le sable d'une plage.
 BLEU: 58.22/100.00 (41.78) | COMET: 68.71/95.68 (26.97)
 Fine-tuning used incorrect in the context "stick out".



English: A man in a blue jacket lighting a cigarette.
 French: Un homme en veste bleue allumant une cigarette.
 Fine-tuning: Un homme en veste bleue **allume** une cigarette.
 Weighting: Un homme en veste bleue allumant une cigarette.
 BLEU: 59.69/100.00 (40.31) | COMET: 92.68/97.31 (4.63)
 Fine-tuning used present simple.



English: A woman putting on a jacket while looking at a clock
 French: Une femme mettant une veste en regardant une horloge
 Fine-tuning: Une femme **en train de** mettre une veste tout en regardant une horloge
 Weighting: Une femme mettant une veste tout en regardant une horloge
 BLEU: 26.13/65.80 (39.67) | COMET: 87.06/94.40 (7.34)
 Fine-tuning used present continuous.

Figure 9: Selected examples with the highest difference in BLEU between Metric-based Loss Weighting and Fine-tuning (Weighting scoring higher) in the MSCOCO testset on the English-to-French language direction of Qwen3-VL 8B. We also report BLEU and COMET scores for the translations by the methods (and their differences), and our description of errors.